capacity retention of more than 94% after 1000 cycles.

During charging, the OH^- ions are oxidised, and oxygen bubbles out on the air electrode. During discharging, oxygen diffuses into the air electrode, and is reduced, generating OH^- ions.

Oxygen evolution reaction (OER) and oxygen reduction reaction (ORR) involving four electrons' transfer are important aspects of Fe-air batteries as shown in Fig. 7.

OER reaction is:

$2H_2O \text{ ------} > O_2 + 4H^+ + 4e^-$

ORR reaction is:

$O_2 + 4H^+ + 4e^- \text{---------} > 2H_2O$

Due to the slow reaction kinetics of electrochemical reactions in Fe-air battery, it is necessary to use a bifunctional catalyst to speed up the reaction during charging and discharging operations. Perovskite families, noble metal groups like Pt, Pd, and Ru prefer the direct $4e^-$ pathway, whereas the indirect two-electron pathway is favoured by carbon materials, transition metals, and oxides.

ORR at air electrode follows the following steps:

1. Oxygen from atmosphere is transferred to the catalyst surface.
2. And then it is absorbed on the active site.
3. Electrons are transferred from the catalyst to oxygen, followed by weakening and removal of the oxygen bond.
4. Finally, the catalyst surface releases the hydroxide ions to the electrolyte.

Design features

To date, most attention has been paid to zinc metal batteries, hoping to capitalise on zinc metal's high capacity of ~820mAh g^{-1} and its low redox potential of -0.76V versus standard hydrogen electrode (SHE). Iron possesses attractive electrochemical properties. Via the two electron transfer, iron theoretically delivers a high specific capacity of ~960mAh g^{-1} and an outstanding volumetric capacity of ~7557mAh cm^{-3}, both of which are higher than zinc. Its redox potential is -0.44, which is 0.3 volts higher than zinc, meaning that its stability in aqueous electrolyte is better. It is also cheap.

All these plus points are the driving force to renew the interest in iron-air batteries. Researchers have now made a viable iron-air battery that is capable of competing with other conventional storage batteries. The exploded view of the iron-air battery developed recently is shown in Fig. 8.

The main features of an iron-air battery consist of each metallic iron anode coupled with two air-breathing electrodes on each side, in a plane parallel configuration. The solid-phase discharge products are deposited on the anode surface and do not transport to the cathode side. The popular approach is to use powdered materials of iron or iron oxides supported by conductive material to fabricate iron electrodes. Polymer-based separators are generally used.

The air electrode is porous and has enhanced electronic conductivity via carbon-based materials, good ionic conductivity, and gas-phase operation. The air electrodes are designed to achieve both the evolution and the reduction of oxygen during the charge and discharge cycles, respectively. The electrode material should be able to withstand the high positive potential during oxygen evolution, must be electrochemically stable, and at the same time able to catalyse the reduction and evolution of oxygen with minimal overpotential losses. Fig. 9(a) shows the sectional arrangement of a single cell, and Fig. 9(b) represents a six-cell configuration. EFY

To be continued...